Marwadi University
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Academic year 24-25
Subject: Probability and Statistics (01CT1401)
G1- A- Tutorial 2 as a part of Term work assessment

Problem Statement:

The Student News Service at Marwadi University (MU) has decided to gather data about the undergraduate students that attend MU. MU creates and distributes a survey of 14 questions and receives responses from 62 undergraduates (stored in the Survey data set).

2.1. For this data, construct the following contingency tables (Keep Gender as row variable)

I am just checking dataset that are this is working or not…
(**I am creating this with using python…**)

```
In [1]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.1.py --wdir
   ID  Gender  Age   Class  ... Spending Computer  Text Messages Wealth
0   1  Female   20  Junior  ...      350   Laptop            200    2.0
1   2    Male   23  Senior  ...      360   Laptop             50   10.0
2   3    Male   21  Junior  ...      600   Laptop            200   70.0
3   4    Male   21  Junior  ...      600   Laptop            250  100.0
4   5    Male   23  Senior  ...      500   Laptop            100    1.0

[5 rows x 15 columns]

In [2]:
```

2.1.1. Gender and Major

Code:

```
import pandas as pd

# Load file for checking:
df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

table1 = pd.crosstab(df['Gender'], df['Major'])
print(table1)
```

```
In [2]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.1.1.py --wdir
Major   Accounting  CIS  ...  Retailing/Marketing  Undecided
Gender                   ...
Female           3    3  ...                    9          0
Male             4    1  ...                    5          3

[2 rows x 8 columns]
```

2.1.2. Gender and Grad Intention

Code:

```
import pandas as pd

# Load file for checking:
df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

table2 = pd.crosstab(df['Gender'], df['Grad Intention'])
print(table2)
```

```
In [3]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.1.2.py --wdir
Grad Intention  No  Undecided  Yes
Gender
Female           9         13   11
Male             3          9   17
```

2.1.3. Gender and Employment

Code:

```
import pandas as pd

# Load file for checking:
df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

table3 = pd.crosstab(df['Gender'], df['Employment'])
print(table3)
```

```
In [4]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.1.3.py --wdir
Employment  Full-Time  Part-Time  Unemployed
Gender
Female              3         24           6
Male                7         19           3
```

2.1.4. Gender and Computer

Code:

```
import pandas as pd

# Load file for checking:
df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

table4 = pd.crosstab(df['Gender'], df['Computer'])
print(table4)
```

```
In [5]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.1.4.py --wdir
Computer  Desktop  Laptop  Tablet
Gender
Female          2      29       2
Male            3      26       0
```

2.2. Assume that the sample is representative of the population of MU. Based on the data, answer the following question:

2.2.1. What is the probability that a randomly selected MU student will be male?

2.2.2. What is the probability that a randomly selected MU student will be female?

```
In [6]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.2.py --wdir
Total students: 62
Male count: 29 Probability Male: 0.4677
Female count: 33 Probability Female: 0.5323
```

2.3. Assume that the sample is representative of the population of MU. Based on the data, answer the following question:

2.3.1. Find the conditional probability of different majors among the male students in MU.

2.3.2 Find the conditional probability of different majors among the female students of MU.

```
In [7]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.3.py --wdir
Cond. Prob. of Major among Male Students:
 Major
Management                0.206897
Retailing/Marketing       0.172414
Other                     0.137931
Economics/Finance         0.137931
Accounting                0.137931
Undecided                 0.103448
International Business    0.068966
CIS                       0.034483
Name: proportion, dtype: float64

Cond. Prob. of Major among Female Students:
 Major
Retailing/Marketing       0.272727
Economics/Finance         0.212121
Management                0.121212
International Business    0.121212
Other                     0.090909
CIS                       0.090909
Accounting                0.090909
Name: proportion, dtype: float64
```

2.4. Assume that the sample is a representative of the population of MU. Based on the data, answer the following question:

2.4.1. Find the probability That a randomly chosen student is a male and intends to graduate.

Code:

```
import pandas as pd

df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

# male and intends to graduate:
male_g_count = ((df['Gender'] == 'Male') & (df['Grad Intention'] == 'Yes')).sum()
total = len(df)

prob_male = male_g_count / total

print("Male & intends to graduate count:", male_g_count)
print("Prob.:", round(prob_male, 4))
```

Output:

```
In [8]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.4.1.py --wdir
Male & intends to graduate count: 17
Prob.: 0.2742
```

2.4.2 Find the probability that a randomly selected student is a female and does NOT have a laptop.

Code:

```
import pandas as pd

df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

# female and does not have a laptop
female_count = ((df['Gender'] == 'Female') & (df['Computer'] != 'Laptop')).sum()
total = len(df)

prob = female_count / total

print("Female & no laptop count:", female_count)
print("Prob.:", round(prob, 4))
```

Output:

```
In [9]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.4.2.py --wdir
Female & no laptop count: 4
Prob.: 0.0645
```

2.5. Assume that the sample is representative of the population of MU. Based on the data, answer the following question:

2.5.1. Find the probability that a randomly chosen student is either a male or has full-time employment?

Code:

```
import pandas as pd

df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

total = len(df)

# count:
male_count = (df['Gender'] == 'Male').sum()
fulltime_count = (df['Employment'] == 'Full-Time').sum()
male_fulltime_count = ((df['Gender'] == 'Male') & (df['Employment'] == 'Full-
Time')).sum()

# prob.
prob = (male_count + fulltime_count - male_fulltime_count) / total

print("male count:", male_count)
```

```
print("full-time count:", fulltime_count)
print("male & full-time count:", male_fulltime_count)
print("prob.:", round(prob, 4))
```

Output:

```
In [10]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.5.1.py --wdir
male count: 29
full-time count: 10
male & full-time count: 7
prob.: 0.5161
```

2.5.2. Find the conditional probability that given a female student is randomly chosen, she is majoring in international business or management.

Code:

```
import pandas as pd

df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

total = len(df)

female_df = df[df['Gender'] == 'Female']
total = len(female_df)

female_ib_count = ((female_df['Major'] == 'International Business') |
                   (female_df['Major'] == 'Management')).sum()

conditional = female_ib_count / total

print("female total:", total)
print("female in IB or manag.:", female_ib_count)
print("Cond. prob.:", round(conditional, 4))
```

Output:

```
In [11]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.5.2.py --wdir
female total: 33
female in IB or manag.: 8
Cond. prob.: 0.2424
```

2.6. Construct a contingency table of Gender and Intent to Graduate at 2 levels (Yes/No). The Undecided students are not considered now and the table is a 2x2 table. Do you think the graduate intention and being female are independent events?

Code:

```
import pandas as pd

df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

# filtering undecided
grad = df[df['Grad Intention'].isin(['Yes','No'])]

# contingency table:
table_grad = pd.crosstab(grad['Gender'], grad['Grad Intention'])
print(table_grad)

# independence check
total = len(grad)
prob_female = (grad['Gender'] == 'Female').sum() / total
prob_yes = (grad['Grad Intention'] == 'Yes').sum() / total
female_and_yes = ((grad['Gender'] == 'Female') & (grad['Grad Intention'] ==
'Yes')).sum() / total

print("P(Female):", round(prob_female,4))
print("P(Yes):", round(prob_yes,4))
print("P(Female AND Yes):", round(female_and_yes,4))
print("P(Female)*P(Yes):", round(prob_female*prob_yes,4))
```

Output:

```
In [14]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.6.py --wdir
Grad Intention  No  Yes
Gender
Female           9   11
Male             3   17
P(Female): 0.5
P(Yes): 0.7
P(Female AND Yes): 0.275
P(Female)*P(Yes): 0.35
```

2.7. Note that there are four numerical (continuous) variables in the data set, GPA, Salary, Spending, and Text Messages.

Answer the following questions based on the data

2.7.1 If a student is chosen randomly, what is the probability that his/her GPA is less than 3?

Code:

```
import pandas as pd

df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

total = len(df)

# count GPA < 3:
gpa_count = (df['GPA'] < 3).sum()
prob = gpa_count / total

print("Students with GPA < 3:", gpa_count)
print("Prob. GPA < 3:", round(prob, 4))
```

Output:

```
In [15]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.7.1.py --wdir
Students with GPA < 3: 17
Prob. GPA < 3: 0.2742
```

2.7.2 Find the conditional probability that a randomly selected male earns 50 or more. Find the conditional probability that a randomly selected female earns 50 or more.

Code:

```
import pandas as pd

df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

male = df[df['Gender'] == 'Male']
female = df[df['Gender'] == 'Female']

# male cond. prob:
male_salary_count = (male['Salary'] >= 50).sum()
prob1 = male_salary_count / len(male)
```

```
# Female cond. prob:
female_salary_count = (female['Salary'] >= 50).sum()
prob2 = female_salary_count / len(female)

print("Male Salary >= 50:", male_salary_count, "Probability:", round(prob1,4))
print("Female Salary >= 50:", female_salary_count, "Probability:", round(prob2,4))
```

Output:

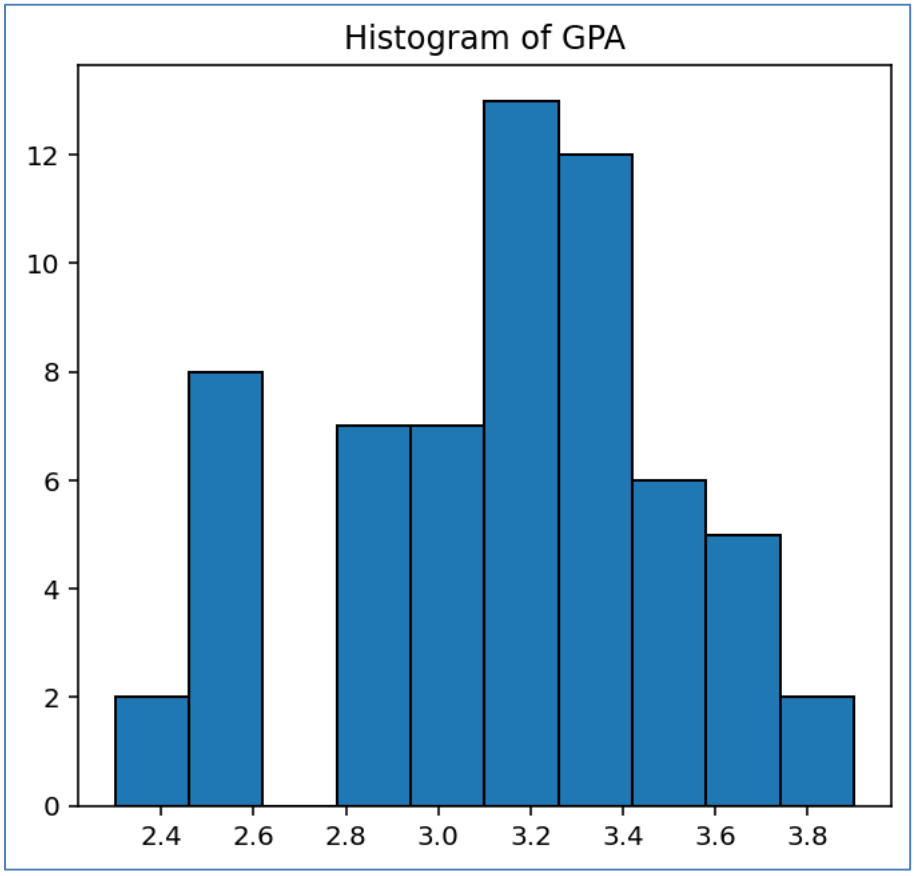
```
In [16]: %runfile C:/Users/Dharmik/COLLEGE/SEM-4/PS/Tutorial-02/2.7.2.py --wdir
Male Salary >= 50: 14 Probability: 0.4828
Female Salary >= 50: 18 Probability: 0.5455
```

2.8. Note that there are four numerical (continuous) variables in the data set, GPA, Salary, Spending, and Text Messages. For each of them comment whether they follow a normal distribution. Write a note summarizing your conclusions.

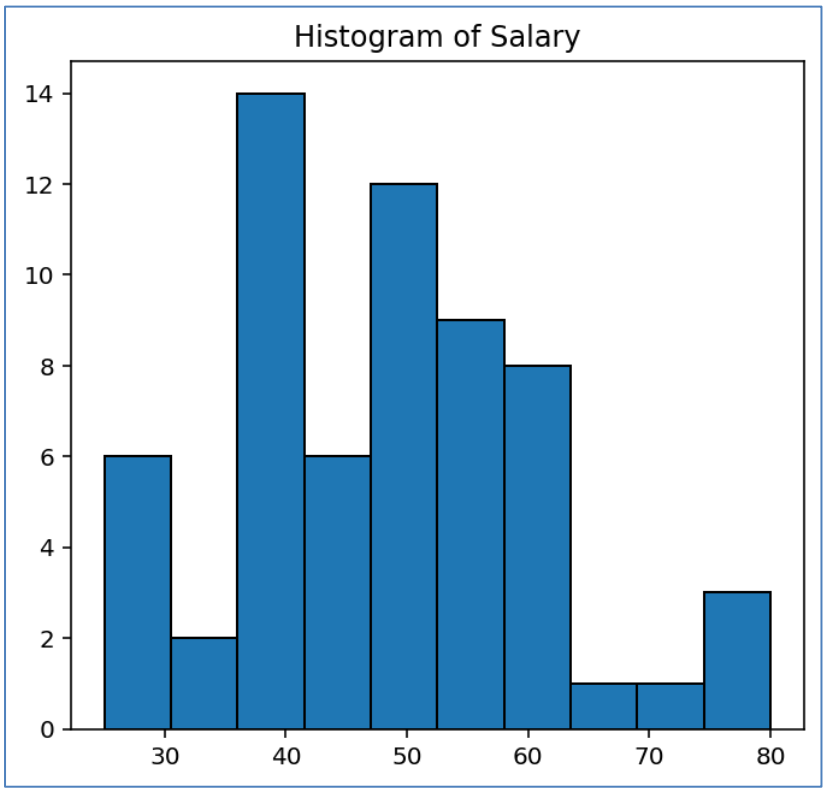
Code:

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_excel("Tutorial 2  G1 A batch_MU_Students.xlsx")

for col in ['GPA','Salary','Spending','Text Messages']:
    plt.figure(figsize=(12,5))

    # histogram:
    plt.subplot(1,2,1)
    plt.hist(df[col], bins=10, edgecolor='black')
    plt.title(f'Histogram of {col}')

    plt.show()
```

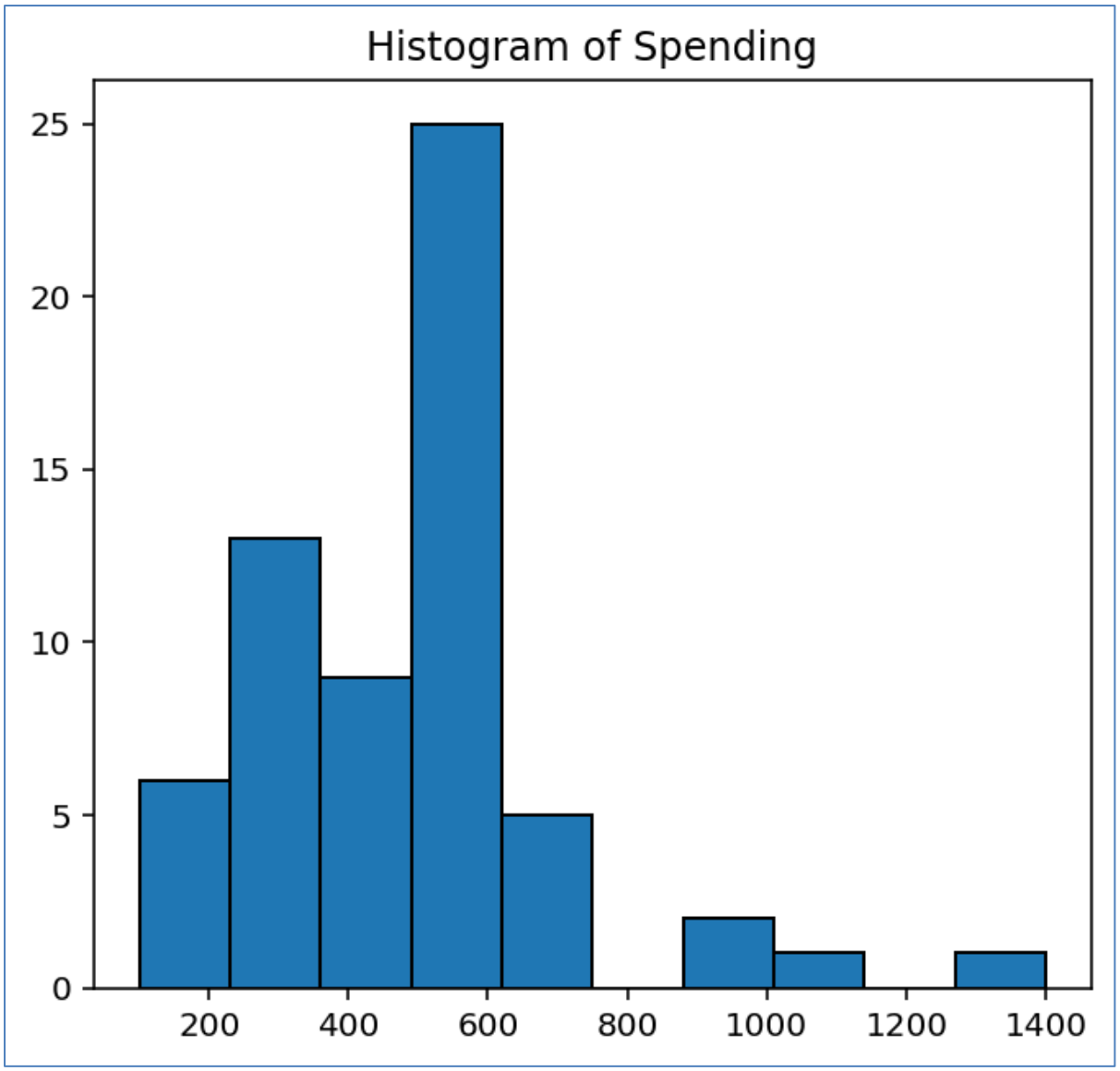
Output:



Conclusion:
The GPA histogram shows most students around 3.2, but the shape is not perfectly bell-shaped. This means GPA does not follow a normal distribution.

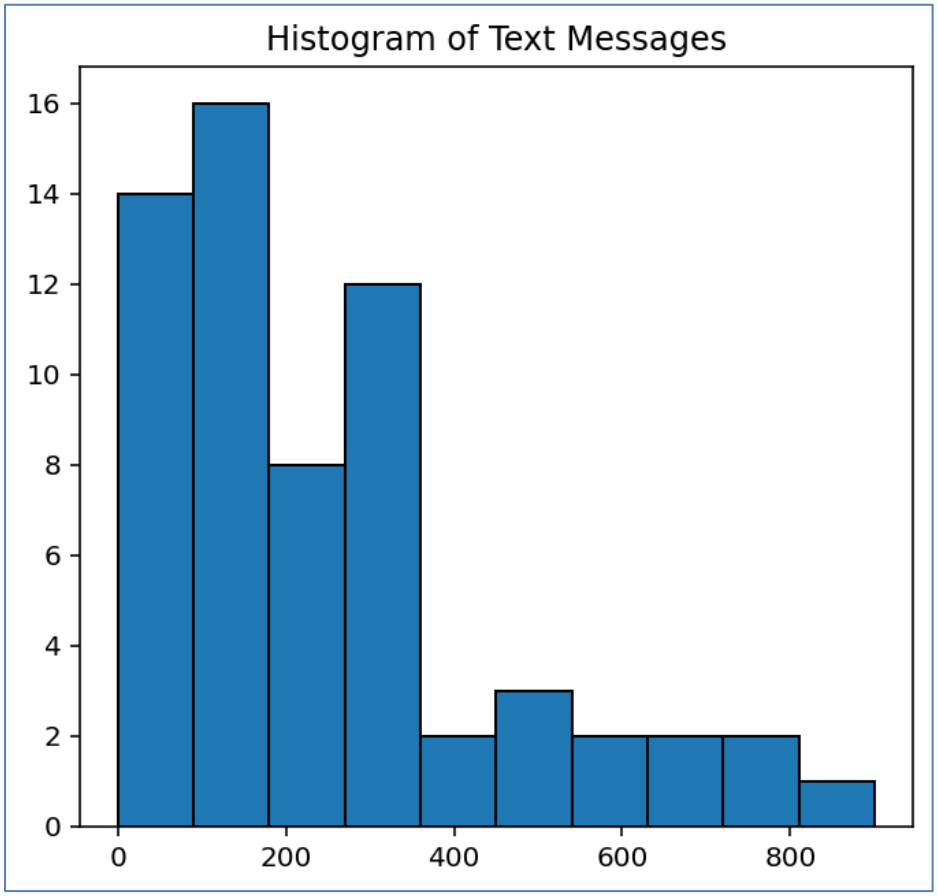


Conclusion:
The salary histogram is skewed to the right with many students earning lower amounts and only a few earning higher. This makes the distribution not normal.



Conlcusion:

Spending values are spread unevenly, with no clear central peak. The histogram suggests that spending habits are not normally distributed.



Conclusion:

Spending values are spread randomly with no clear central peak. The histogram suggests that spending habits are not normally distributed.